

# Nearby Vibes

## Implementation Roadmap

a phased build plan for Claude Code

**Companion to:** Nearby Vibes MVP PRD v1.3

**Purpose:** Build sequencing, dependencies, and phase exit criteria

**Prepared for:** Engineering build-out via Claude

July 21, 2026

# Contents

- How to use this roadmap . . . . . 2
- Dependency overview . . . . . 2
- Phase 0 — Foundation & Environment . . . . . 3
- Phase 1 — Accounts, Auth & Venue Onboarding . . . . . 3
- Phase 2 — Post / Special Lifecycle & Media Pipeline . . . . . 3
- Phase 3 — Discovery Experience (Tile/Map, Age Gate, Location) . . . . . 4
- Phase 4 — Follow System & Engagement Scoring . . . . . 4
- Phase 5 — Push Notification Engine . . . . . 4
- Phase 6 — QR Flow, Geofence & Arrival Confirmation . . . . . 5
- Phase 7 — Analytics & Dashboards . . . . . 5
- Phase 8 — Security, Content & SEO Hardening . . . . . 5
- Phase 9 — Performance, Load Test & Launch Readiness . . . . . 6
- Cross-cutting notes . . . . . 6

## How to use this roadmap

The PRD (v1.3) specifies *what* to build. This document specifies *in what order* — the dependency chain that determines which phase blocks which, so a coding agent doesn't build push notifications before there's anything to notify about, or a dashboard before there's an event stream feeding it.

**Recommended first move:** give a coding agent both the PRD and this roadmap, and ask it to confirm or challenge the phase plan before writing any code — not to start executing Phase 0 immediately. If it disagrees with a sequencing call here, that's worth surfacing before work starts, not after.

Each phase below lists: its goal, what it depends on, the PRD sections it implements, and the exit criteria that mark it done — mostly drawn directly from the PRD's own Acceptance Criteria (§8), so "done" isn't a judgment call.

## Dependency overview

graph TD

```

P0[Phase 0: Foundation & Environment]
P1[Phase 1: Accounts, Auth & Venue Onboarding]
P2[Phase 2: Post Lifecycle & Media Pipeline]
P3[Phase 3: Discovery Experience]
P4[Phase 4: Follow System & Engagement Scoring]
P5[Phase 5: Push Notification Engine]
P6[Phase 6: QR Flow, Geofence & Arrival]
P7[Phase 7: Analytics & Dashboards]
P8[Phase 8: Security, Content & SEO Hardening]
P9[Phase 9: Performance, Load Test & Launch]

P0 --> P1
P1 --> P2
P2 --> P3
P1 --> P3
P3 --> P4
P1 --> P4
P2 --> P5
P4 --> P5
P1 --> P6
P2 --> P6
P4 --> P6
P0 -.event stream starts here, feeds forward.-> P7
P3 --> P7
P5 --> P7
P6 --> P7
P7 --> P8
P8 --> P9

```

Note the dotted line: analytics event logging (post\_impression, post\_open, navigation\_click, qr\_scans, etc., §14.3) should be instrumented incrementally as each phase ships the action it logs — not bolted on retroactively in Phase 7. Phase 7 builds the *dashboards*; the *events* need to already be flowing by then. Visual design execution (the companion Design Brief) runs in parallel starting now, feeding into Phases 1, 2, 3, and 6 as those screens get built.

---

## Phase 0 — Foundation & Environment

**Goal:** a deployable skeleton with no features yet, but every account, service, and schema the rest of the build depends on.

**Depends on:** nothing — this is the starting point.

**Build:**

- Provision accounts: Google Cloud (Places API + Maps API, billing attached), Firebase (Auth + Cloud Messaging), Cloudflare (R2, Turnstile, Bot Management, Rate Limiting, DNS/WAF), Mapbox, DigitalOcean (Managed Postgres with PostGIS, app hosting), Vercel, domain + cdn./api. subdomains (§20.1, §20.2).
- Repo scaffold: Next.js frontend, NestJS backend (confirm monorepo vs. split repos before starting).
- CI pipeline: lint, type-check, unit tests, dependency vulnerability scan, build check on every PR; auto-deploy to staging on merge to main (§20.3).
- Database: create every core table from the data model up front, even for features that land in later phases — users, venues, venue\_staff, posts, media\_assets, follows, devices, push\_events, qr\_scans, arrival\_confirmations, reports, static\_pages / static\_page\_versions, admin\_audit\_log, and the analytics event stream (§14.2, §14.3). Enable PostGIS, add spatial (GiST) indexes per §19.5.
- .env.example covering every credential named above.

**Exit criteria:** empty frontend and backend deploy to staging; a health-check endpoint responds; migrations run clean against the managed Postgres instance; CI is green on a trivial PR.

---

## Phase 1 — Accounts, Auth & Venue Onboarding

**Goal:** every account type can be created, verified, and a venue can be claimed.

**Depends on:** Phase 0.

**Build:** REQ-ACC-001-012 and 030-034 (personal/business signup, Google/Apple/email auth, password policy, email verification gate, 30-day “remain signed in” cap, Turnstile, Social Media Manager invite + restricted role), REQ-ONB-010-021 (Google Places claim search, map confirmation, 1:1 Place-ID enforcement, fraud rules §10.5, business email domain blocklist §10.4, venue profile fields). Reference: §7.1–7.2, §12.1 sequence, §13.1–13.2 API, §18.1–18.3 security/RBAC.

**Exit criteria:** §8.1 and §8.2 acceptance criteria pass — business signup rejects free-email domains, unverified email blocks onboarding, sessions expire at 30 days, duplicate Place-ID claims are rejected, a second venue claim from the same business account is rejected.

---

## Phase 2 — Post / Special Lifecycle & Media Pipeline

**Goal:** a venue can create, schedule, publish, and archive a post, with working media upload/processing.

**Depends on:** Phase 1 (need a claimed, profile-complete venue to post from).

**Build:** REQ-POST-001-032 (Draft → Scheduled → Active → Archived lifecycle, one-active-post-per-venue rule, 7-day max duration, recurring windows, immutable drink disclaimer, reporting), REQ-MEDIA-001-013 (upload validation, HEIC conversion, 9:16 canvas + crop, aspect-ratio auto-formatting, one-time Sharp compression pipeline, CDN delivery). Reference: §7.4-7.5, §11.1 state diagram, §12.2 sequence, §13.3 API, §10.2-10.3 business rules.

**Exit criteria:** §8.3 acceptance criteria pass — second Active post blocked, Scheduled→Active→Archived transitions fire on time, Archived posts can't be deleted, drink disclaimer can't be edited or removed.

---

## Phase 3 — Discovery Experience (Tile/Map, Age Gate, Location)

**Goal:** users can actually find and view live specials nearby. This is where the Design Brief's Home Tile/Map screens get built for real.

**Depends on:** Phase 2 (needs posts to discover), Phase 1 (needs venues).

**Build:** REQ-UX-001-042 (Tile View, Map View, marker states/clustering, PWA install prompt, cookie consent, fuzzy-location fallback, WCAG 2.1 AA), age gating per §10.1 (blur + inline gate, food never gated). Reference: §7.3, §9.2-9.5, §9.11, §13.4 API.

**Exit criteria:** §8.4 and §8.5 acceptance criteria pass — drink tiles render blurred alongside ungated food tiles, denied location returns fuzzy results with the benefit-explainer prompt, map markers render correct category states, live-count banner is accurate.

---

## Phase 4 — Follow System & Engagement Scoring

**Goal:** trial follows form, accumulate engagement points, and convert to hard follows on schedule.

**Depends on:** Phase 3 (post views/expands are scoring events), Phase 1 (accounts to attach follows to).

**Build:** the confirmed scoring model — Push Open 1, Post View 1, Post Expand 2, Directions Click 3, Return Session 3, Attendance 3 — evaluated on a 2-hour batch cycle, 5-point conversion threshold, 48-hour trial expiry. Reference: §10.6, §11.2 state diagram, follows table (§14.2).

**Exit criteria:** §8.6 acceptance criteria pass — trial follow created on qualifying trigger, conversion fires once 5 points accumulate within 48h (checked on the 2-hour cycle, not instantly), trial expires cleanly if the threshold isn't met, an already-following user sees the correct message rather than a duplicate prompt.

---

## Phase 5 — Push Notification Engine

**Goal:** the full notification pipeline — discovery pushes, follow pushes, bundling, caps, quiet hours, operational pushes to venues.

**Depends on:** Phase 2 (posts trigger pushes), Phase 4 (follow state gates the follow-push path), Phase 0 (Firebase Cloud Messaging wiring).

**Build:** REQ-PUSH-001-021 in full — 15-minute bundling window, bundling priority order, 60-minute minimum interval, 2/24h and 8/7-day device caps, 2-posts-per-venue-per-day cap, quiet hours (12-8 AM venue-local, held not dropped), multi-day post fires once at activation,

venue-owner “post went live” / “post ended” operational pushes. Reference: §15 in full, §11.4 state diagram, §12.3 sequence, §13.7 API.

**Exit criteria:** §8.7 acceptance criteria pass — bundling produces one push for multiple qualifying posts, device caps block a push once exceeded, quiet-hours pushes are held and delivered at 8 AM rather than dropped, the expiry-reminder push respects the 9-minute new-post suppression window.

---

## Phase 6 — QR Flow, Geofence & Arrival Confirmation

**Goal:** the universal QR code resolves correctly via geofence, and arrival confirmation works independently of it.

**Depends on:** Phase 1 (venue geofence data must exist from onboarding), Phase 2 (posts to display), Phase 4 (QR scans can create/convert follows).

**Build:** REQ-QR-001-006 — single universal code ([nearbyvibes.com/qr](https://nearbyvibes.com/qr)), three-tier distance resolution ( $\leq 30\text{m}$  exact /  $30\text{--}100\text{m}$  nearby fallback /  $>100\text{m}$  out of range), multi-venue tie-break/disambiguation, `qr_scans` logging across all tiers. Arrival confirmation per §10.8 ( $30\text{m}$  geofence + 4-hour post-view window), independent of QR. Reference: §16 in full, §12.4–12.5 sequences, §13.6 API.

**Exit criteria:** §8.8 acceptance criteria pass — all three distance tiers render correctly, overlapping venues trigger disambiguation rather than a wrong guess, arrival confirmation fires correctly on its own  $30\text{m}/4\text{h}$  rule regardless of QR involvement.

---

## Phase 7 — Analytics & Dashboards

**Goal:** the venue dashboard and admin dashboard, reading from an event stream that’s already been flowing since Phase 1.

**Depends on:** Phase 0 (schema), and functionally on Phases 3, 5, 6 (the events being displayed need to already be logged by then).

**Build:** venue dashboard — “Customers Brought In” headline metric plus the five capped visuals (Post Views Over Time, Push Performance funnel, Post Engagement, Visits from Posts, Post Performance Card), Social-Manager-scoped view. Admin dashboard — network health, trending venues (reusing the §10.6 engagement score), push/QR health, growth, platform/PWA metrics. Reference: §17 in full, §13.8 API, Metabase wiring per §20.1.

**Exit criteria:** §8.11 acceptance criteria pass — headline metric is the most prominent element on the venue dashboard, admin network totals are visible without further navigation.

---

## Phase 8 — Security, Content & SEO Hardening

**Goal:** everything that isn’t a user-facing feature but is a launch blocker.

**Depends on:** the feature surface being largely complete (Phases 1–7), since this phase audits it.

**Build:** full security baseline pass (§18 — JWT/session handling, RBAC enforcement server-side, rate limiting on auth/claim/report endpoints, input validation, secure token storage, dependency scan clean), static/legal pages populated with the ready-to-paste

Terms/Privacy/AUP/disclaimer copy (§7.7), SEO implementation (REQ-SEO-001-006 — structured URLs, schema.org markup, sitemap, Search Console submission), full WCAG 2.1 AA audit against real screens (including the confirmed brand palette’s contrast, §9.12).

**Exit criteria:** §8.10 reporting acceptance criteria pass; structured data validates against schema.org; accessibility audit finds no AA-blocking issues; CI’s dependency scan is clean.

---

## Phase 9 — Performance, Load Test & Launch Readiness

**Goal:** prove the system holds at target scale, and the launch process itself is rehearsed.

**Depends on:** Phase 8.

**Build:** load test at 10,000 concurrent users / 200 venues against staging (§19.6), monitoring/alerting on the core discovery/post-read/post-publish paths (§20.4), automated daily backups with a rehearsed restore drill (§20.5), Manhattan-only rollout plan executed (§20.6).

**Exit criteria:** the full §8.12 MVP “done” gate — all six conditions — verified end-to-end in staging before production launch: nearby discovery with correct age-gating, push delivery, native-maps navigation hand-off, arrival confirmation, QR-driven follow states, and the full attribution funnel visible on the venue dashboard.

---

## Cross-cutting notes

- **Analytics instrumentation is not a phase, it’s a habit.** Log the relevant event (post\_impression, post\_open, navigation\_click, qr\_scans, push\_sent/push\_opened, arrival\_confirmation) in the same PR that ships the feature generating it, not later. Phase 7 will be blocked or produce empty dashboards if this slips.
- **Design execution runs in parallel, not sequentially.** The companion Design Brief can be executed against starting now; its output should land in Phases 1 (auth/onboarding screens), 2 (post wizard), 3 (Tile/Map — the bulk of the surface), 6 (QR landing), and 7 (dashboards) as those phases come up, rather than waiting for a single “design phase.”
- **No open decisions remain.** All five items from the PRD’s Appendix A are confirmed (QR scheme, engagement scoring, legal copy, map provider, visual direction) — nothing in this roadmap should be blocked on a pending Product decision.